

THE RATTY2TAILS PAPERS · THE MACHINE SERIES
· NO. II



The Swarm and the Stone

Starlink, Starmind, and a sketch of how a planet of autonomous
agents might come to settle its accounts in Bitcoin

Written under the name **Ratty2Tails**
September 2026

The Swarm and the Stone

Starlink, Starmind, and a sketch of how a planet of autonomous agents might come to settle its accounts in Bitcoin

TL;DR

- SpaceX is quietly building two constellations, not one. **Starlink** moves bits between points on Earth. **Starmind** — a separate megaconstellation announced through 2026, built with Nvidia's Rubin and Vera silicon and filed with the FCC for up to a million satellites — is meant to run AI compute in orbit and beam down only the answers.
- Moving inference off the ground turns two of the scarcest inputs to today's AI build-out — land and continuous power — into things low Earth orbit has for free: sun that never sets in the right orbit, and a vacuum that helps rather than hinders cooling.
- Cheap, abundant, location-agnostic compute is the precondition for a shift already underway: from a few chatbots serving billions of humans, to billions of small, persistent AI agents transacting continuously with each other, with no human clicking "buy."
- Machine-to-machine commerce breaks the assumptions correspondent banking was built on — legal personhood, business hours, chargebacks, KYC. It needs a settlement rail an agent can hold and use without asking anyone's permission. Bitcoin, and specifically the Lightning Network, is the only mature candidate built for exactly that.
- Sketch: agents meter and pay each other in satoshis for compute, bandwidth and data in real time, with Bitcoin's base layer as final settlement — playing something like the role a gold vault once played under a network of clearing banks.
- The same orbital reach that could hand someone with no fibre, no bank branch and no data centre a dish and a Lightning wallet could also let a handful of orbital-compute operators capture the next round of first-mover monetary gains. Every leapfrog in this essay has a historical precedent that rewarded the platform owner as often as the periphery it reached.
- Everything past the first section is informed speculation about a real, current, unresolved piece of infrastructure — flagged as such throughout, not reported as settled fact.

A note on method: Section 1 sets out what is publicly known and verifiable as of September 2026. Everything from Section 2 onward is Ratty's own scenario-building — an attempt to reason forward from real, current infrastructure decisions to their plausible monetary consequences. Treat it as informed speculation, not as reporting.

§ 0 Introduction and Scope

There is a habit, in writing about the future of money, of assuming the next monetary order will be built by central bankers, or by regulators, or at the very least by someone who filed a white paper. This essay is a bet that the more important decisions are currently being made by rocket engineers, and that almost nobody discussing "the future of Bitcoin" or "the future of AI" has put the two conversations in the same room.

In the summer of 2026, SpaceX confirmed it is building a second satellite constellation, entirely distinct from Starlink, dedicated not to moving data between two points on Earth but to computing on it in orbit. It named this constellation Starmind, filed with the U.S. Federal Communications Commission for authority to operate up to a million of these satellites, and announced a hardware partnership with Nvidia to fit them with the same class of silicon — Rubin GPUs, Vera CPUs — that powers the largest terrestrial data centres on the planet. First prototype launches are pencilled in for 2027.

Nobody, including SpaceX, has demonstrated that this is economically sound. Amazon's own cloud chief has publicly doubted the unit economics of a million-satellite data centre. This essay takes no position on whether Starmind specifically succeeds. What it takes seriously is the underlying pressure Starmind is a response to: terrestrial AI compute is running into physical walls — land, grid interconnects, cooling water, permitting timelines — that orbit simply does not have. Somebody will eventually solve for that pressure, whether it is SpaceX or a competitor drawing the same conclusion five years later.

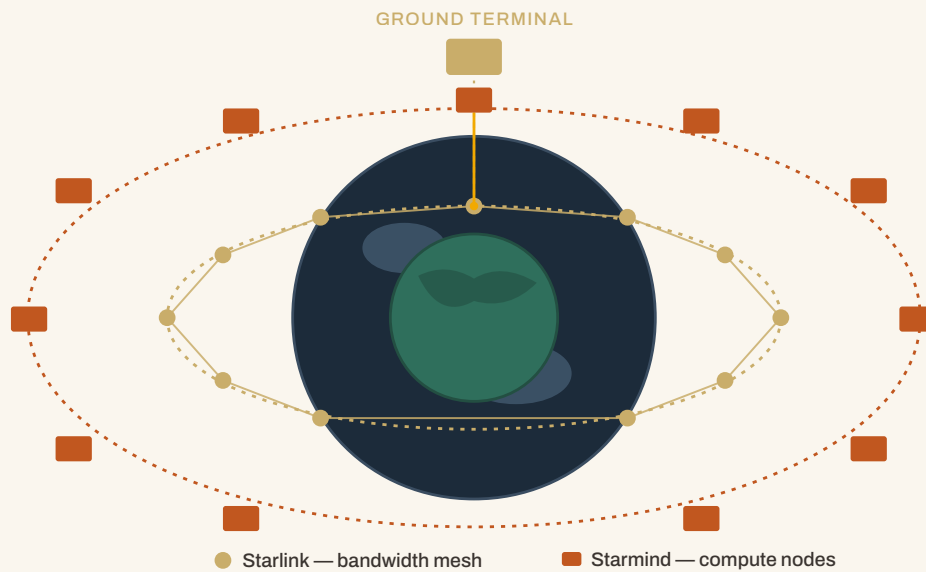
The essay is built in three movements. First, the facts: what Starlink and Starmind actually are, and how they differ (§1). Second, a chain of reasoning about what abundant, location-agnostic compute does to the economics of AI agents, and why agents transacting with other agents need a settlement layer that no bank, card network or fiat currency was built to provide (§§2–4). Third, a deliberately imaginative sketch of how Bitcoin could end up serving as that settlement layer, what it would do to the shape of the global economy, and — because this is where these essays usually fall short — what it would mean, for better and for worse, in the parts of the world with no fibre, no data centre and often no functioning central bank to begin with (§§5–7).

Two real satellite programmes, a well-documented physics constraint, and one already-existing piece of orbital Bitcoin infrastructure are the entire factual scaffolding here. Everything built on top of that scaffolding is Ratty sketching, not Ratty reporting.

§ 1 Two Constellations

Starlink needs no introduction to most readers of this series: a broadband relay network in low Earth orbit, now numbering in excess of ten thousand active satellites, providing low-latency internet to places fibre has never reached and, increasingly, to phones directly. It is, in essence, a very fast pipe. Data goes up from one point on Earth, crosses a mesh of laser inter-satellite links, and comes down at another point on Earth. The satellites themselves do not think about the data; they move it.

Starwind is a different animal entirely, and the distinction matters more than SpaceX's habit of stellar branding might suggest. Where Starlink is plumbing, Starwind is meant to be a processor. Each Starwind satellite is, in Elon Musk's own description on SpaceX's 2026 earnings call, essentially an orbital rack of Nvidia's NVL72 system — the same cabinet-scale, liquid-cooled compute block increasingly used to train and run the largest AI models on the ground, minus the ground. The published specifications for the first-generation "AI1" satellite describe a spacecraft roughly 20 to 30 metres tall with wingspan-scale solar arrays generating on the order of 200 kilowatts, feeding a compute payload rated at over 100 kilowatts, cooled by more than 100 square metres of deployable liquid radiator panels. SpaceX filed with the FCC in January 2026 for authority to launch and operate up to one million of these, flown in sun-synchronous orbits chosen specifically to maximise continuous daylight, and linked back to Earth via the existing Starlink optical mesh. Two prototypes are expected in early 2027, with volume production targeted for later that year at a dedicated facility in Bastrop County, Texas.



Starlink relays bits between points on Earth. Starmind computes on them in orbit, then beams down results.

Figure 1. Starlink and Starmind occupy the same orbital neighbourhood but do fundamentally different jobs — one moves information, the other transforms it.

It is worth being honest about how far ahead of itself this all is. As of writing, Starmind exists as an FCC filing, a hardware partnership announcement, and a pair of prototypes not yet launched. AWS's own leadership has questioned publicly whether a million-satellite compute constellation is "economically viable" at all, and there are serious unresolved questions about orbital debris, thermal management at scale, and whether laser downlink bandwidth can actually carry the volume of inference traffic such a constellation would generate. None of that is this essay's concern to adjudicate. What matters for the argument that follows is narrower and more defensible: **the physical logic behind Starmind is sound even if the specific programme fails.** Terrestrial data centres are bumping into land availability, grid interconnection queues that now run years long in some markets, and local water tables strained by cooling demand. Low Earth orbit has none of those constraints. It has permanent sun in the right orbit, the vacuum of space as a (admittedly awkward) heat sink, and no neighbours to object to the noise. Somebody, eventually, builds the orbital data centre. The only real question is when, and who.

A PRECEDENT ALREADY IN ORBIT

Bitcoin has, in fact, already been broadcast from space for nearly a decade. Blockstream Satellite, launched in 2017 and still operating today, beams the Bitcoin blockchain continuously to most of the globe from a small existing fleet of commercial

satellites, letting anyone with a modest dish run a full node with no internet connection at all, and — since a later upgrade — send and receive Lightning payments the same way. It is a fraction of the scale this essay imagines, but it is the existence proof that Bitcoin and orbital infrastructure are not strangers to one another.

§ 2 Why This Multiplies Agents, Not Just Chatbots

The current AI economy is still shaped, in large part, like the mainframe era: a small number of very large, very expensive facilities, timeshared by an enormous number of human users typing into a chat window. That shape is a consequence of scarcity — compute is expensive enough that it has to be pooled and rationed, so the natural unit of the industry is "one model serving millions of people," not "one small agent working continuously on behalf of one person or one task."

Cheap, abundant, always-on compute erodes that logic. It has happened before: when computing moved from time-shared mainframes to the personal computer, the natural unit stopped being "one machine per building" and became "one machine per desk." When it moved again, from the desktop to the cloud and the smartphone, the natural unit became "one small always-connected device per pocket, mostly idle, occasionally busy." Each step down in the cost of a marginal compute-hour multiplied the number of independent, persistent things doing computation, rather than simply making the existing big machines cheaper.

There is good reason to expect the same pattern with AI agents specifically, independent of whether Starmind itself succeeds. An agent that can hold context, take actions, and run continuously — watching a shipping fleet's maintenance schedule, rebalancing a small trading book, negotiating spot prices for freight capacity, monitoring a farm's irrigation against weather forecasts — is far more useful running all the time than running only when a human remembers to open an app. But "running all the time" only makes economic sense once the marginal cost of the compute it consumes falls well below the value of the small, continuous decisions it makes. Orbital compute, if it delivers on its own economics, is one candidate for getting there; falling chip costs and more efficient model architectures are others. The direction of travel does not depend on any single vendor.

The consequence worth dwelling on is this: a world of a few large models serving billions of humans is a world where the interesting transactions are almost all human-to-machine — a person paying a subscription, a business paying an API bill. A world of billions of small, continuously running agents is a world where an enormous and rapidly growing share of transactions become **machine-to-machine**: one agent buying ten milliseconds of inference from another, one agent paying a satellite operator

for bandwidth, one agent selling a forecast to another agent a thousand times a second. That shift in who is transacting is the hinge the rest of this essay turns on.

§ 3 The Missing Rail — Why Machines Can't Bank

Every payment rail in wide use today — card networks, ACH, SWIFT, the correspondent banking system that ultimately clears almost all of it — was designed around a small set of assumptions about the parties using it. They are legal persons who can be identified. They transact in round-ish amounts, infrequently enough that a human can review each one. They tolerate settlement measured in hours or days. They accept that a bank, somewhere, can freeze, reverse, or simply decline to process the payment.

None of those assumptions survive contact with a machine economy. An autonomous agent has no legal personhood to put on a KYC form. It may need to pay another agent a fraction of a cent, ten thousand times an hour, for slivers of inference or bandwidth metered in milliseconds — a transaction pattern for which the marginal cost of card-network interchange alone would exceed the value being exchanged many times over. It cannot wait until Monday for a wire to clear. And critically, it often has no natural counterparty it can trust with custody: two agents, owned by two different, possibly adversarial firms, negotiating a price for compute in real time, have no shared bank and no reason to extend each other credit.

What the old rails assume

- A legal identity behind every account
- Settlement measured in hours to days
- Transactions large enough to absorb fixed fees
- A trusted intermediary holding the ledger
- Business hours, human review, chargebacks

What a machine economy needs

- A bearer instrument any process can hold
- Final settlement in seconds, not days
- Fees that don't dominate sub-cent transfers
- No permission required to send or receive
- Uptime around the clock, with no counterparty

This is not a hypothetical gap waiting for a clever fintech to fill it with a slightly faster API on top of the same banking rails. The requirement is structural: something that behaves like cash — final, bearer, no intermediary needed to authorise the transfer — but that also settles at the speed and denomination of a machine, over whatever link happens to be available, including a satellite one with no live connection back to a correspondent bank in London or New York.

§ 4 Why Bitcoin, Specifically

Readers of the Monetary History series will recognise the shape of the argument this series has made about Bitcoin for humans: fixed, disinflationary issuance removes the first-receiver advantage that Cantillon identified in every monetary expansion since the Mississippi Company, because no single actor — not a central bank, not a fleet operator, not a satellite constellation's parent company — can create new units and spend them before the price adjusts. What is new here is that the same property matters even more once the holders of the asset are machines rather than people.

An AI agent that earns and holds its own working capital needs a monetary asset no counterparty can debase out from under it while it isn't looking, because it has no legal recourse, no lawyer, and often no human even monitoring the balance in real time. A fixed-supply bearer asset solves that by construction. Self-custody solves the second half of the problem: an agent can hold its own private keys exactly the way it already holds its own model weights or API credentials — no custodian, no account-opening process, no jurisdiction whose banking regulator needs to approve a non-human counterparty.

The Lightning Network is the part of the stack that makes this practical at machine speed rather than merely possible in principle. A payment channel between two agents settles in a fraction of a second, at a cost of a small fraction of a cent, with no third party in the loop once the channel is open — properties that map almost exactly onto the requirements laid out in the previous section. Lightning already supports the kind of continuous, streamed micropayment a machine economy would generate — metering a data feed sat by sat rather than billing for it in a monthly invoice — because the same low-friction settlement that makes sense for a bar tab in sats makes even more sense for a thousand ten-millisecond slices of GPU time.

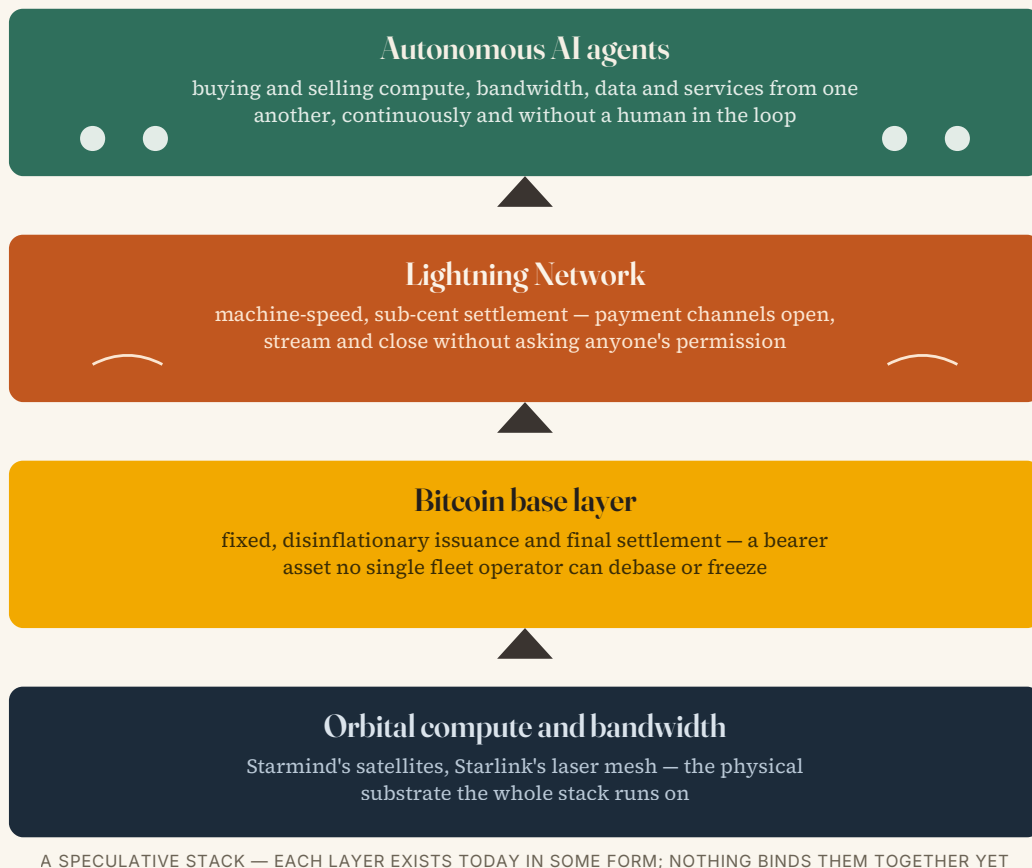


Figure 2. A speculative settlement stack. Every layer already exists in some form today; nothing currently binds them together into the whole.

None of this requires that Bitcoin displace the dollar, the pound, or the yuan for human retail spending — an argument this series has never made and does not make here. It requires something narrower and, in some ways, more plausible: that when a machine has to choose a settlement asset for a counterparty it does not trust and cannot sue, the properties that make Bitcoin awkward for a weekly shop — volatility, irreversibility, the need to self-custody keys — stop being drawbacks and start being exactly what is required.

§ 5 Sketching the Mechanism

Here is where the essay stops describing anything that exists and starts imagining how it might come together. Take it as a scenario, not a roadmap.

Picture an orbital compute market in which Starmind-class satellites — whoever eventually operates them — sell inference and storage by the millisecond to any agent that can reach them over a laser or radio downlink. Pricing that fine-grained cannot be metered and invoiced in the traditional sense; there is no back office fast enough or cheap enough to bill in fiat at that resolution. Instead, each agent opens a Lightning channel with the compute provider — or, more likely, with an intermediary node that aggregates many small buyers the way a Lightning Service Provider already aggregates retail users today — and streams satoshis continuously as it consumes cycles, closing out and rebalancing the channel periodically against the Bitcoin base layer.

Widen the lens and the same mechanism generalises. A logistics agent pays a weather-data agent a few sats for a forecast used in a single routing decision. A trading agent pays an orbital-relay agent for the lowest-latency path to an exchange. A content-moderation agent rents burst capacity from an idle compute agent during a traffic spike, and pays it back automatically the moment the spike passes. None of these payments individually is interesting. In aggregate, they describe an economy that runs on a settlement asset few of its participants ever "choose" in any deliberate sense — it is simply the asset the infrastructure defaults to, the way TCP/IP was never voted on by the people who now depend on it.

Bitcoin does not need to win an argument about being better money for people. It only needs to be the money the machines already agree on, because a shared foundation of trustless settlement is worth more to two mutually suspicious pieces of software than a slightly better exchange rate would be.

Two caveats belong here, in the spirit of the epistemic honesty this series tries to keep. First, stablecoins are the obvious, less ideologically loaded competitor for exactly this niche, and it would be dishonest to pretend they aren't currently winning

more of the actual machine-payment pilots being built in 2026. This essay's Austrian bias suggests that a dollar-denominated stablecoin merely relocates the Cantillon problem — whoever issues it can still, in principle, freeze or dilute it — but a fair-minded reader should treat that as a real, contested Section-5-only argument rather than a settled fact. Second, Bitcoin's price volatility is a genuine obstacle to its use as a unit of account for agents that need to reason about costs in stable terms; the more plausible near-term role is as a settlement and reserve layer beneath a stack of faster-moving units of account, in something like the way gold once sat beneath a system of banknotes it never itself circulated as.

§ 6 Effects on the Global Economy

If even a fraction of this comes to pass, the effects ripple well beyond payments plumbing. A meaningful share of global economic activity — logistics routing, content moderation, financial execution, agricultural planning — would come to run on infrastructure owned by a very small number of firms with the capital to launch and operate orbital compute at all. That is not a new story; it is the same story told about telegraph cables, transatlantic fibre, and cloud computing before it, each of which concentrated a surprising amount of durable economic advantage in the handful of firms that owned the pipe.

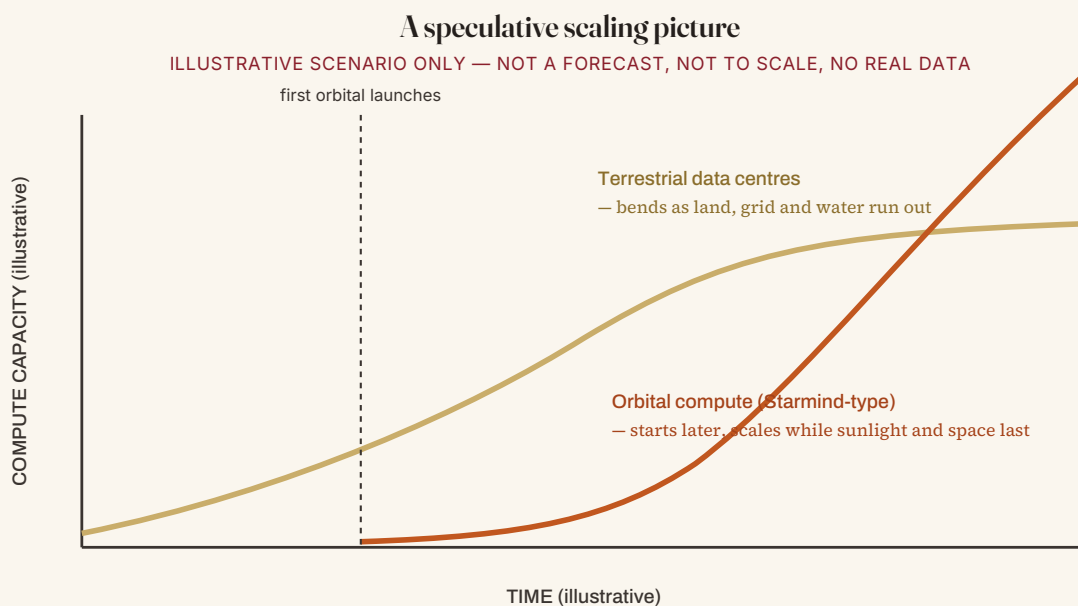


Figure 3. An entirely illustrative sketch of the shape of the argument — terrestrial compute bending against physical ceilings while an orbital alternative, if it works, keeps climbing. No real figures are implied.

Bitcoin's role in this picture is genuinely double-edged, and the honest version of this essay has to sit with both edges rather than resolving them into a tidy conclusion. On one hand, a neutral, fixed-supply settlement asset is the one part of this stack that cannot be captured by whichever firm wins the race to build the compute layer — a Starmind-operator monopoly on orbital inference is still a monopoly even if its customers pay it in Bitcoin, but at least that monopoly cannot also print the money its

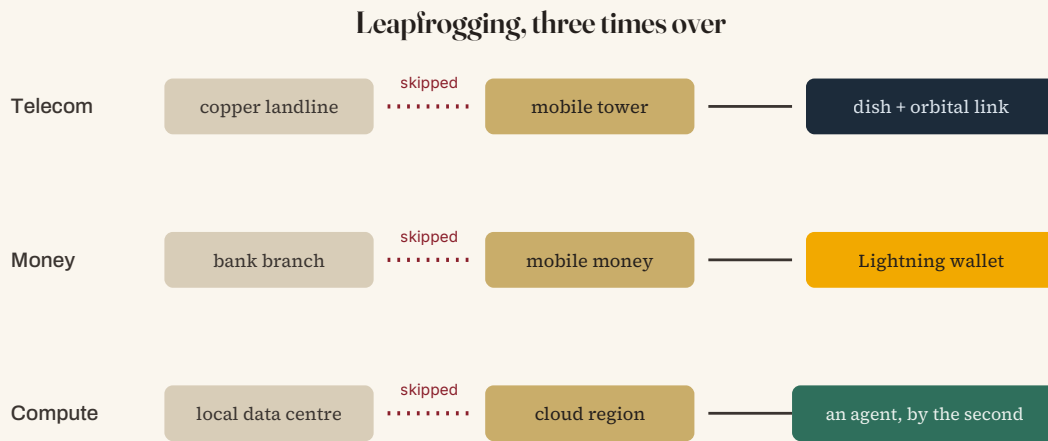
customers are forced to hold, the way a company town's scrip once could. On the other hand, the earliest and largest holders of the new machine economy's working capital will very plausibly be the same capital-rich firms that can afford to launch satellites in the first place — a first-receiver advantage in the underlying compute market that Bitcoin's monetary neutrality does nothing to touch, because the advantage sits one layer below the currency, in who owns the machines earning it.

Put bluntly: sound money at the settlement layer does not guarantee a fair distribution of the economy sitting on top of it. It guarantees only that, whatever the distribution turns out to be, nobody can quietly re-slice it after the fact by debasing the unit everyone agreed to settle in.

§ 7 Not Forgetting the Underdeveloped World

It would be easy, and this series has criticised other writers for exactly this, to sketch a future built on satellites and machine agents and never once ask what any of it means for the roughly two billion people who today have no bank account, patchy or absent grid electricity, and no fibre within reach. This essay's own back catalogue — The Dark Continent, on the Berlin Conference's carve-up of Africa and the long shadow it casts over the continent's monetary sovereignty — is a standing reminder that infrastructure built by rich-world capital to serve rich-world ends has a poor track record of accidentally serving everyone else.

The optimistic case is real, and worth stating plainly before the caution that follows it. Mobile money in East Africa did not wait for a bank branch to reach every village; it leapfrogged the branch entirely, riding infrastructure — the mobile tower — that was built for an unrelated reason and happened to reach further and cheaper than the alternative it replaced. A dish pointed at a Starlink-and-Starlink sky does the same thing to the data centre and the bank branch simultaneously: a smallholder farmer, a rural clinic, or a small trading business could in principle access both AI-agent services and Bitcoin's settlement layer without a single mile of fibre or a single local bank ever being built. For a region where the average cost of remitting money is still measured in double-digit percentages, and where a substantial share of any given country's own currency has, at some point in living memory, been debased by its own central bank, both halves of that offer — cheap machine intelligence and a monetary asset no local authority can inflate away — are not small things.



Each row needed less fixed infrastructure on the ground than the one before it.
Whether that keeps favouring the periphery, or just relocates the toll booth, is the open question.

Figure 4. The same leapfrogging pattern, repeated in telecom, then in money, and — this essay speculates — next in access to intelligence itself.

The caution is just as real, and this series would be doing its readers a disservice to leave it out. Every leapfrog in that diagram still required someone to own the tower, the wallet infrastructure, or the satellite constellation — and ownership of the new toll booth has, historically, migrated no further from rich-world capital than it had to. A Starmind-class satellite is not a village well; it is a piece of aerospace infrastructure requiring billions of dollars, an FCC filing, and access to launch capacity that, as of 2026, exactly one company on Earth possesses at the necessary scale. A farmer with a dish and a Lightning wallet is meaningfully better off than a farmer with neither — but she is also now a customer, indefinitely, of whoever owns the satellites, in a relationship with no local regulator, no local competitor, and no local recourse if the price of a millisecond of inference simply rises. That is not a reason to romanticise the status quo she started from, which usually offered no bank branch and no data centre either. It is a reason to hold the leapfrog story and the extraction story in the same hand, the way this series tried to with the Berlin Conference, rather than choosing whichever one makes the technology sound better.

Bitcoin's part in this is, again, genuinely narrow and genuinely useful rather than a solution to the whole problem. It cannot make the satellite operator competitive, or force better terms, or substitute for the domestic institutions — property rights, contract enforcement, a functioning court — that ultimately determine whether any of this compounds into durable local prosperity or merely a more efficient extraction pipe. What it can do is guarantee that whatever value does accrue locally — a farmer's sats earned selling weather data to a logistics agent, a clinic's sats earned running an

idle compute node overnight — cannot be quietly inflated away by a domestic central bank the way remittance income and savings so often have been. That is a real and specific thing to offer the periphery of this story. It is not the same thing as offering it the centre.

§ 8 The Skeptic's Case

A fair essay owes its argument a serious opponent, so here is the case against most of the above coming to pass, at least on any timeline worth planning around.

- **The economics may simply not work.** Launching, powering, cooling and replacing a million satellites is an enormous and continuous capital outlay, and AWS's leadership has already voiced the obvious objection publicly. Orbital compute may end up a niche complement to terrestrial data centres rather than a replacement for them.
- **Regulatory and physical limits are real.** The FCC has not approved the million-satellite filing as of writing, and even a fraction of that number raises legitimate orbital-debris and Kessler-syndrome concerns that regulators, insurers, and other satellite operators will not wave through lightly.
- **Stablecoins, not Bitcoin, may win the machine-payment niche.** They are already further along in actual pilots, denominated in a unit machines and their human owners find easier to reason about, and backed by firms with every incentive to make them the default rail before Bitcoin's Lightning Network reaches comparable liquidity and tooling.
- **Volatility is a real cost, not just an ideological objection.** An agent managing a working-capital balance in an asset that can move ten percent in a day faces genuine operational risk that a well-run treasury function, human or machine, cannot simply wave away.
- **Concentration may simply reproduce itself.** If the same three or four capital-rich firms end up owning the satellites, the compute, and the largest Lightning nodes routing the payments, a "neutral settlement layer" changes less about the distribution of power than its neutrality suggests.

None of these objections make the underlying pressure — terrestrial compute's physical ceiling, and machine-to-machine commerce's mismatch with correspondent banking — go away. They are reasons to expect a slower, messier, more contested version of this essay's sketch than the tidy stack in Figure 2 suggests, and reasons this piece is filed as speculation rather than forecast.

§ 9 Summary

IN SHORT

SpaceX is building two very different constellations under similar names: Starlink moves data, Starmind — still unlaunched, still economically unproven, but backed by real hardware partnerships and a real FCC filing — is meant to compute on it in orbit, sidestepping the land, power and cooling limits now constraining terrestrial AI infrastructure. If that or something like it succeeds, it plausibly accelerates a shift already underway from a few large models serving humans to a proliferation of small, continuously running AI agents transacting with each other. Machine-to-machine commerce at that speed and denomination has no natural home in a banking system built for legal persons and business hours. Bitcoin's fixed supply and bearer-asset properties, married to the Lightning Network's machine-speed settlement, are the closest thing that currently exists to infrastructure built for exactly that problem — not because Bitcoin will replace the dollar for people, but because it may simply be the asset machines default to settling in. The same dynamic that could hand the settlement layer of a new machine economy to the technologically peripheral world, exactly as mobile money once did, could just as easily hand its ownership to the same small set of capital-rich firms that own the satellites. Both outcomes are compatible with everything described here; which one dominates is a question of ownership and policy, not of the technology's inherent tendencies.

A METHODOLOGICAL NOTE — Section 1 of this essay is a factual account of public, verifiable information as of September 2026, sourced below. Sections 2 through 8 are Ratty's own speculative reasoning about how that infrastructure could plausibly develop and what it might mean, offered in the spirit of informed scenario-building rather than prediction. Nothing in Sections 2–8 should be read as reporting on any company's actual plans for Bitcoin integration, which — as far as is publicly known — do not currently exist.

§ 10 Sources & Further Reading

- [1] Space.com, "Another 'Star' is born: SpaceX names AI megaconstellation 'Starlink'" (June 2026) — on the naming and stated scale of the Starlink constellation relative to Starlink.
- [2] Reporting on SpaceX's August 2026 earnings call and Nvidia hardware partnership announcement for the Starlink AI1 compute payload, including Rubin GPU / Vera CPU specifications and the Bastrop County, Texas "Gigasat" production facility.
- [3] Coverage of SpaceX's January 2026 FCC filing for up to one million Starlink satellites at 500–2,000 km altitude, and of AWS leadership's public skepticism about the economics of orbital data centres at that scale.
- [4] Basenor / Sawyer Merritt reporting on the Starlink AI1 satellite's published specifications (compute payload, solar array output, thermal system, orbit altitude) from SpaceX's Starlink website, July 2026.
- [5] Blockstream, official documentation for Blockstream Satellite, launched 2017 and operating continuously since, broadcasting the Bitcoin blockchain and supporting Lightning transactions without a terrestrial internet connection.
- [6] Prior essays in this house's Monetary History series — The Ledger and the Stone; The Cantillon Effect; The Stone That Cannot Be Quarried; The Dark Continent — for the Austrian monetary framing, the first-receiver / Cantillon mechanism, and the Berlin Conference parallel drawn on in §§4, 6 and 7.